PRE-ALGEBRA (8TH) · UNIT 5: PATTERNS AND BIVARIATE DATA

Screen Time & Sleep Study

Use scatter plots, lines of fit, and two-way tables to study screen time and sleep.

CP2

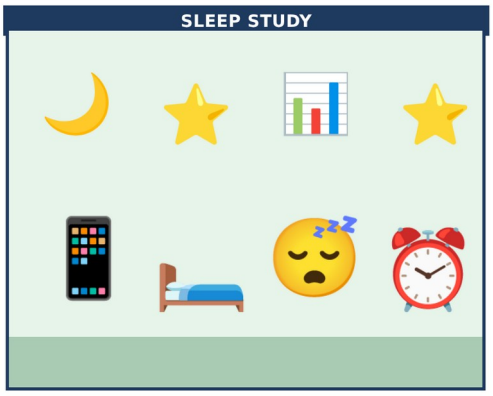
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

The school health committee wonders whether screen time is connected to how much students sleep. They collected data from students and want a clear, honest report.

You will analyze the data with scatter plots, a line of best fit, and a two-way table, and present what the data can and cannot tell us.



Model

sleep ≈ −0.45 × screen + 9.4

(hours per night)

Survey

50 students

phone in bedroom? sleeps 8+ h?

Learning Goals

- I can make a scatter plot and describe the association.
- I can use a line of best fit to make a prediction.
- I can read a two-way table and compare percents.

Project Timeline

When	What you will do
Day 1	Scatter Plot
Day 2	The Line of Fit
Day 3	Two-Way Table
Final	Research Poster + reflection

Part 1: Scatter Plot

Day 1 · Use the Study Data card.

Study Data

Screen time (h) → Sleep (h)
1 → 9.0 · 2 → 8.6 · 2.5 → 8.4
3 → 8.0 · 4 → 7.6 · 4.5 → 7.5
5 → 7.0 · 6 → 6.6 · 6.5 → 6.4 · 8 → 5.8

Two-way table (50 students)
Phone in bedroom: 6 sleep 8+ h, 24 sleep < 8 h
No phone in bedroom: 14 sleep 8+ h, 6 sleep < 8 h

1. Make a scatter plot with screen time on the x-axis and sleep on the y-axis. Describe the association.

   Starter: *As screen time goes up, sleep ___*

2. A new student reports 8 hours of screen time and 9 hours of sleep. Is this point an outlier? Explain.

3. Use the line sleep ≈ −0.45 × screen + 9.4 to predict sleep for 7 hours of screen time.

Part 2: The Line of Fit

Day 2 · Use y = −0.45x + 9.4.

4. Predict the sleep for 3.5 hours of screen time.

5. What does the slope −0.45 mean?

6. What does the y-intercept 9.4 mean?

Part 3: Two-Way Table

Day 3 · Use the survey of 50 students.

7. How many students have a phone in the bedroom? How many sleep 8 or more hours?

8. What percent of students with a phone in the bedroom sleep 8+ hours? What percent of students without one?

9. Is there an association between a phone in the bedroom and sleeping 8+ hours? Explain.

Final Product: Research Poster

Your final product must include:

- Scatter plot with the line of best fit
- Two predictions with the line
- Slope and intercept explained in words
- Two-way table with percents
- A research poster with one clear finding

Reflection

R1. What does a negative association look like on a scatter plot?

R2. Why compare percents instead of counts in the two-way table?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Scatter plot, model, predictions, and two-way analysis are accurate and correctly interpreted. Goes beyond the minimum with a creative, realistic choice.	Scatter plot, model, predictions, and two-way analysis are accurate and correctly interpreted.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Poster states findings clearly and explains limits (association vs. causation, extrapolation). Polished, organized, and visually engaging.	Poster states findings clearly and explains limits (association vs. causation, extrapolation). Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.